VTPEH 6105

Lecture 09

Hypothesis Testing (and Predictions): Multivariate Models

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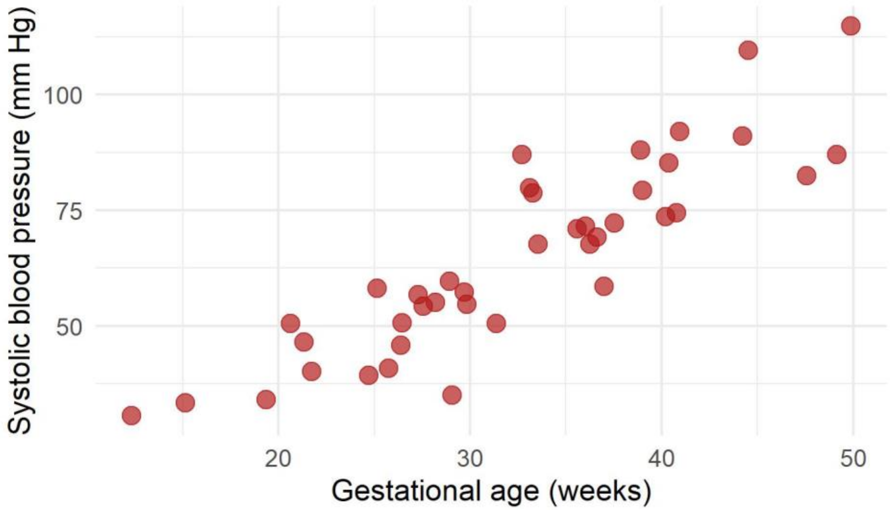
Linear Regression Refresher

Linear regressions: estimation of the parameters by OLS

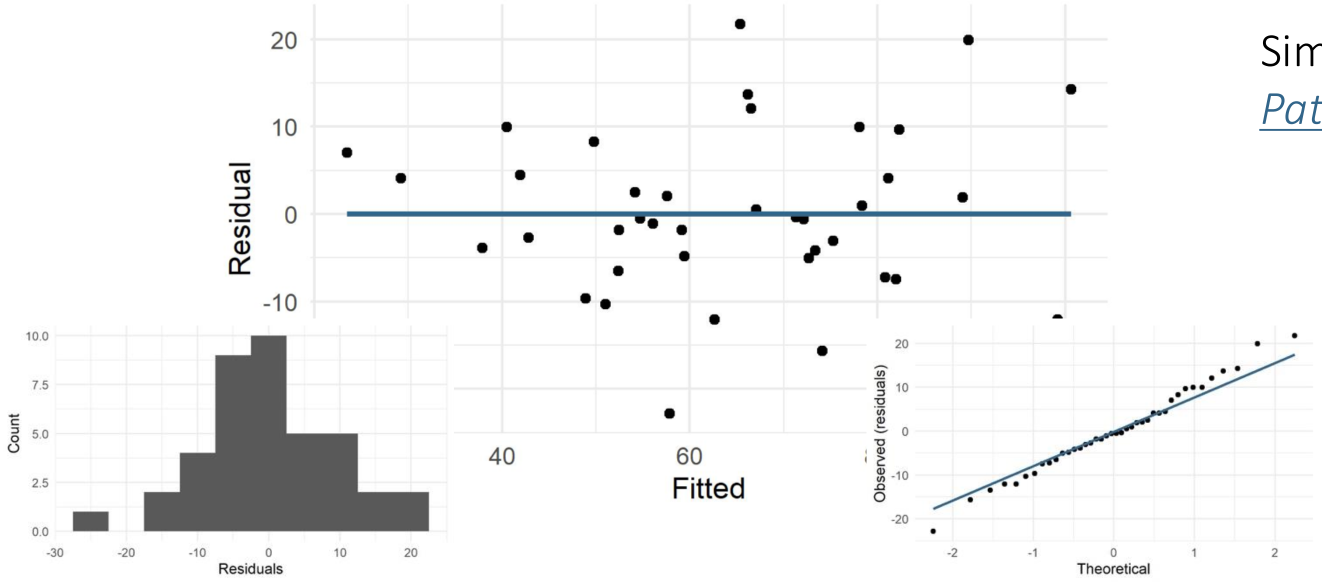
- On the raw data…



Simulated data inspired from *Avery's Neonatology: Pathophysiology and Management of the Newborn*

Linear regressions: assumption testing, final validation

- On the **residuals**…



Simulated data inspired from *Avery's Neonatology: Pathophysiology and Management of the Newborn*

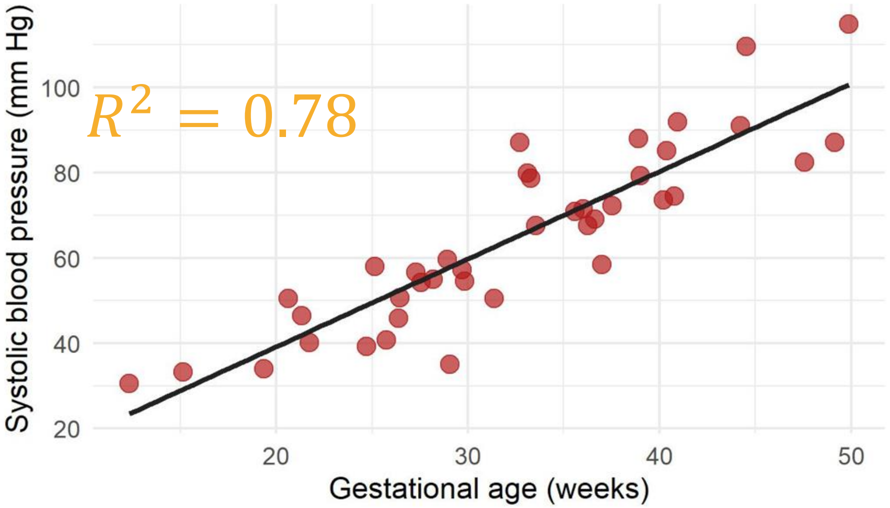
Linearity — Residuals centered on 0

Normality — Normal distribution of the residuals

Homoscedasticity — Uniform dispersion of the residuals around 0

Linear regressions: assumption testing, final validation

- Model is validated, we can move to the interpretation…



Simulated data inspired from *Avery's Neonatology: Pathophysiology and Management of the Newborn*

$$y_i = 2.06\, x_i + 1.88 + \varepsilon_i$$

Effect size

$y_i - \hat{y}_i$

Observed – Fitted (or predicted)

$\neq 0$

We look for the parameters a and b that best describe $y_i = ax_i + b + \varepsilon_i$

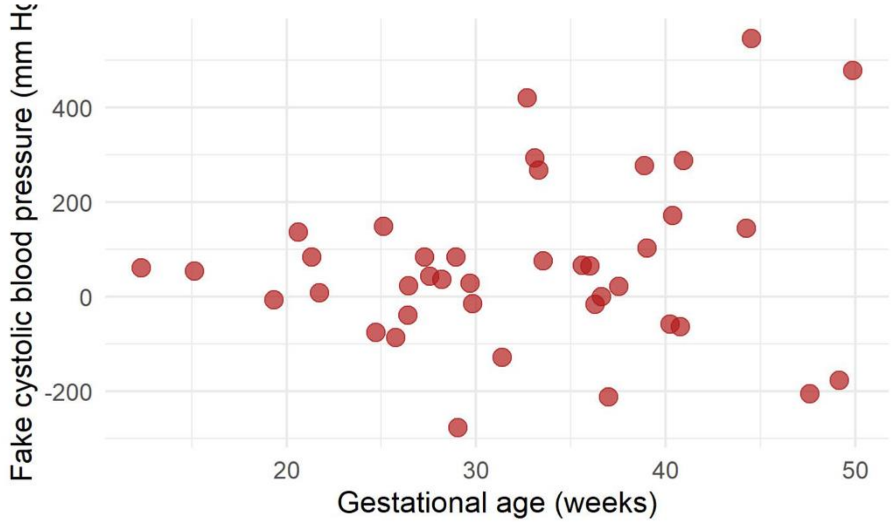
Hand calculation, or `lm(y ~ x, data)` in R

Slope $\hat{a} = 2.06 \pm 0.34$ and intercept $\hat{b} = 1.88 \pm 11.42$

Linear regressions: assumption testing

~~Ordinary~~ Generalized Least Squares

- Another example… With less clean data



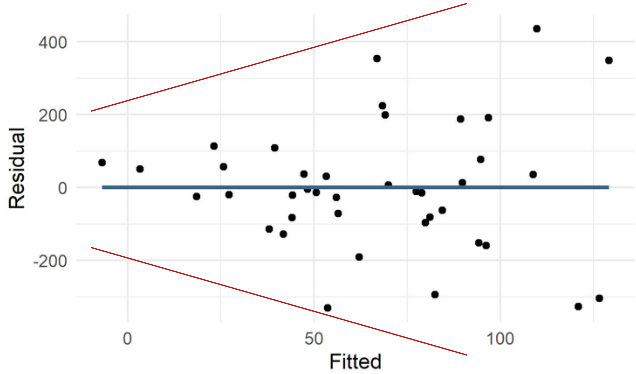
Simulated data, assuming flu is a seasonal with high incidence in winter

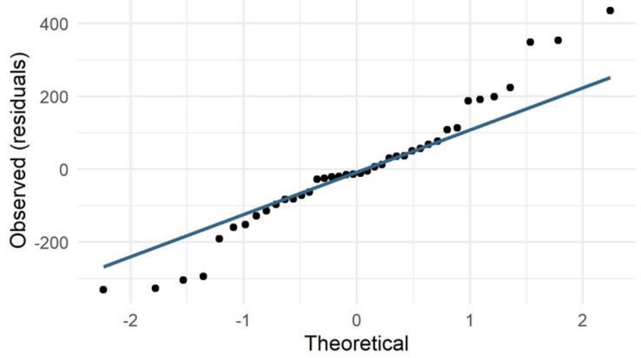
Slope $\hat{a} = -0.06 \pm 0.02$ and intercept $\hat{b} = 29.14 \pm 3.27$
$R^2 = 0.58$

Linearity?

Normality?

Homoscedasticity?

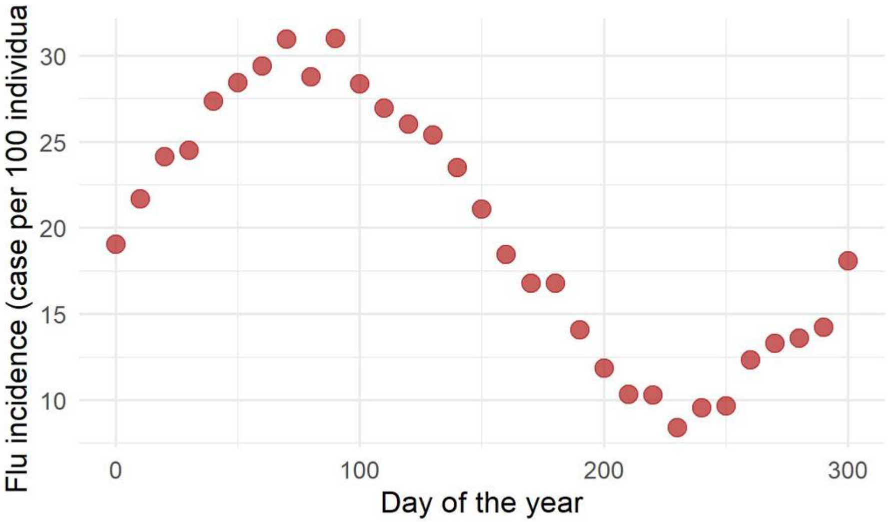




Linear regressions: assumption testing

A linear model is irrelevant (a mechanistic model would be more appropriate)

- Another example… With a non-linear relationship



Simulated data, assuming flu is a seasonal with high incidence in winter

Slope $\hat{a} = 0.78 \pm 0.19$ and intercept $\hat{b} = -0.11 \pm 3.97$

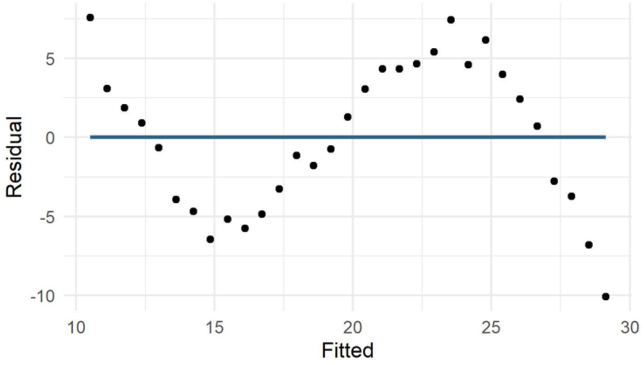
$R^2 = 0.58$

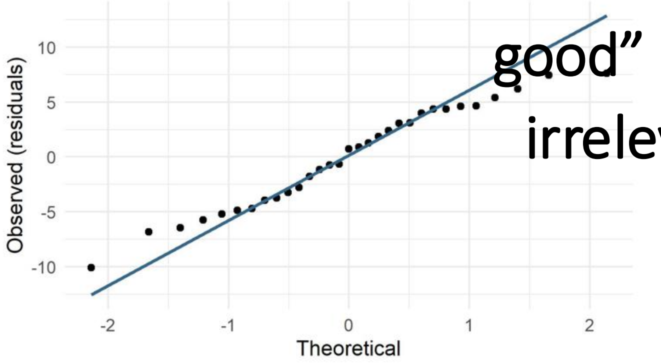
The linear model still runs and might "look good" but is irrelevant

Linearity?

Normality?

Homoscedasticity?



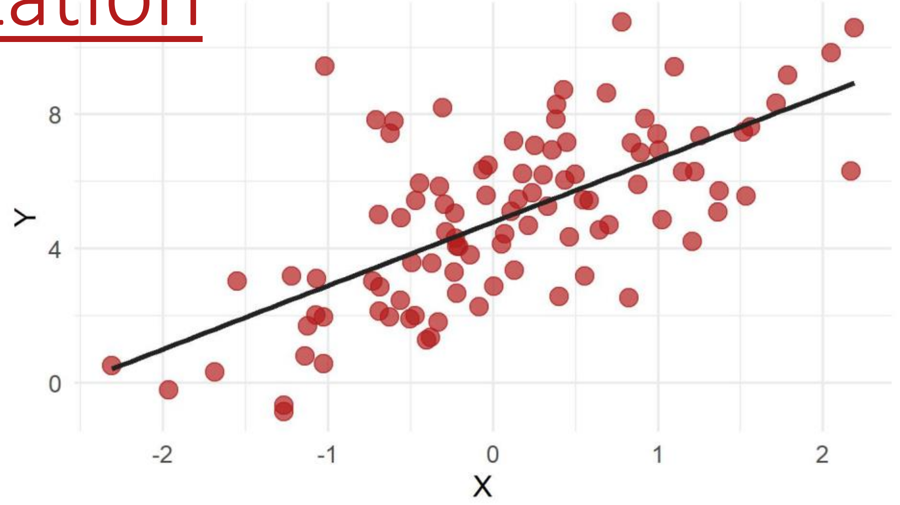


Multivariate Models

Multivariate linear regressions: formalization

- Regression: **estimation the relationships** between variables

$$Y \sim X \text{ or } Y \sim X_1, X_2, \dots, X_k$$



→ An objective way to **predict** or estimate the value of one variable, given a value of another variable

- **Univariate linear** model
- **Multivariate linear** model

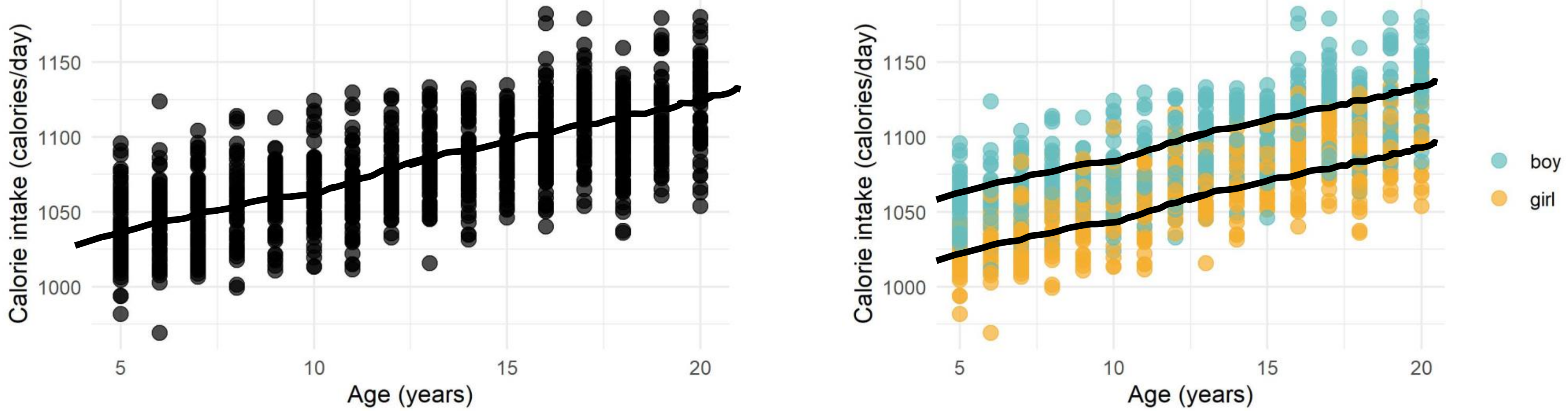
$$Y = a\,X + b \text{ and } y_i = \underbrace{a\,x_i + b}_{\text{Predicted value}} + \varepsilon_i$$

Individual variation, or noise: deviance from the predicted value

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k \text{ and}$$
$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon_i$$

Multivariate linear regressions: formalization

- Example: calorie intake in kids



- **Univariate linear** model $Y = a\,X + b$ and $y_i = a\,x_i + b + \varepsilon_i$
- **Multivariate linear** model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$ and

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon_i$$

Multivariate linear regressions: <u>assumption</u> testing

- **Independence**: Y values are statistically independent (via ε_i)

$$y_i = a\,x_i + b + \varepsilon_i$$

- **Linearity between X and Y**: the means of the subpopulations of Y all lie on the same straight line defined by

$$\mu_{y|x} = a\,x + b$$

- **Normality**: <u>for each value of X</u> there is a subpopulation of Y values; these subpopulations must be normally distributed
- **Homoscedasticity**: the variances of the subpopulations of Y are all equal and denoted by σ^2

Study design

Step 0 – **Biological assumption**
Step 1 – Visual exploration of the **raw data**
Step 2 – Visual exploration of the **model residuals**

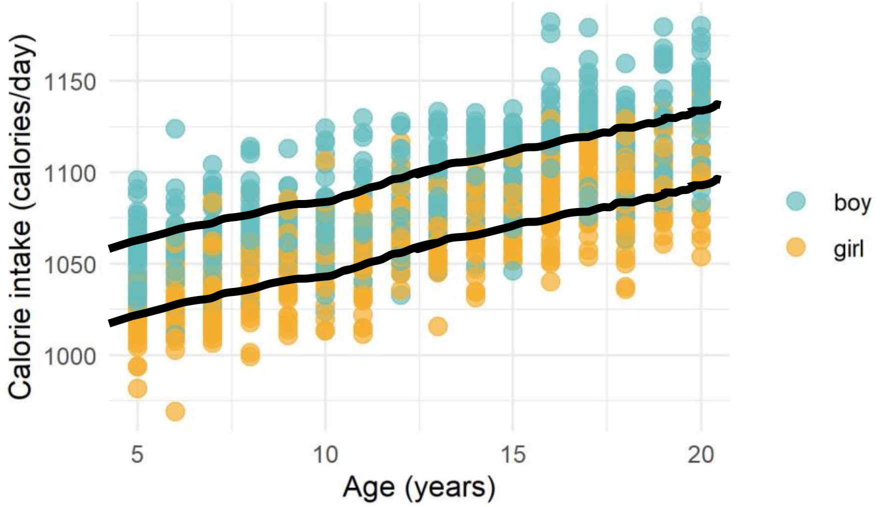
Multivariate linear regressions: assumptions

- Example: calorie intake in kids

Linearity?

Normality?

Homoscedasticity?



- **Multivariate linear** model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$ and $y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon_i$

Multivariate linear regressions: assumptions

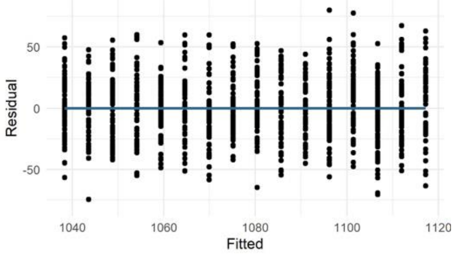
$$R^2 = \frac{SSY - SSE}{SSY}$$

- Example: calorie intake in kids

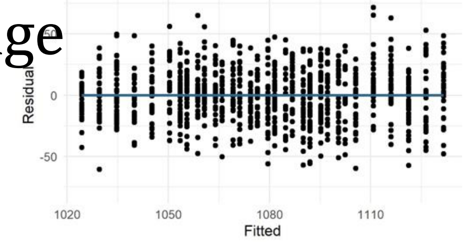
$$R^2_{\text{adj}} = 1 - \frac{SSE/_{n-\boldsymbol{k}-1}}{SSY/_{n-1}}$$

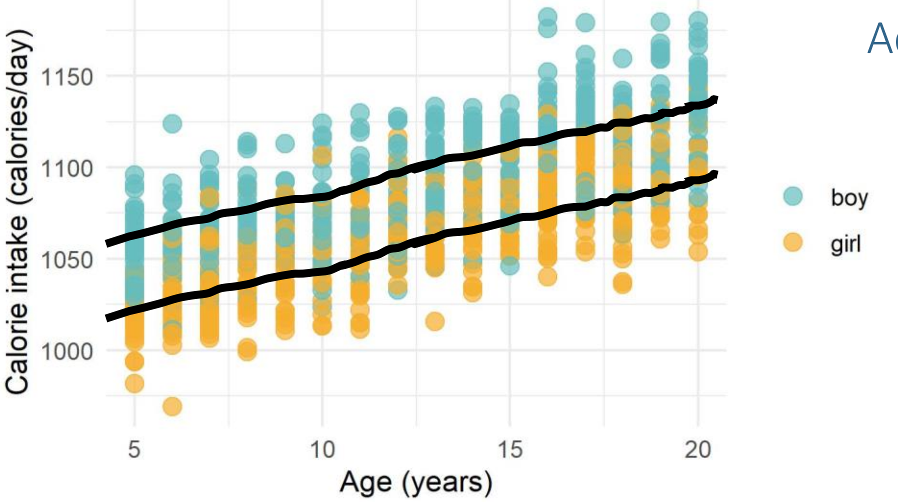
Adjusted to the number of covariates

$$\text{Cal} = \beta_0 + \beta_1 \text{Age}$$
$$R^2_{\text{adj}} = 0.47$$



$$\text{Cal} = \beta_0 + \beta_1 \text{Gender} + \beta_2 \text{Age}$$
$$R^2_{\text{adj}} = 0.63$$





Gender

Age

- **Multivariate linear** model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k \text{ and}$$
$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon_i$$

Adding More Variables…

Multivariate linear regressions: formalization

$$R^2 = \frac{SSY - SSE}{SSY}$$

- Example: calorie intake in kids

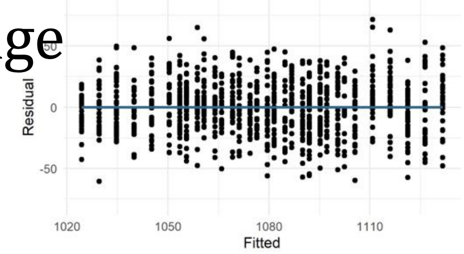
$$R^2{}_{\text{adj}} = 1 - \frac{SSE/_{n-\boldsymbol{k}-1}}{SSY/_{n-1}}$$

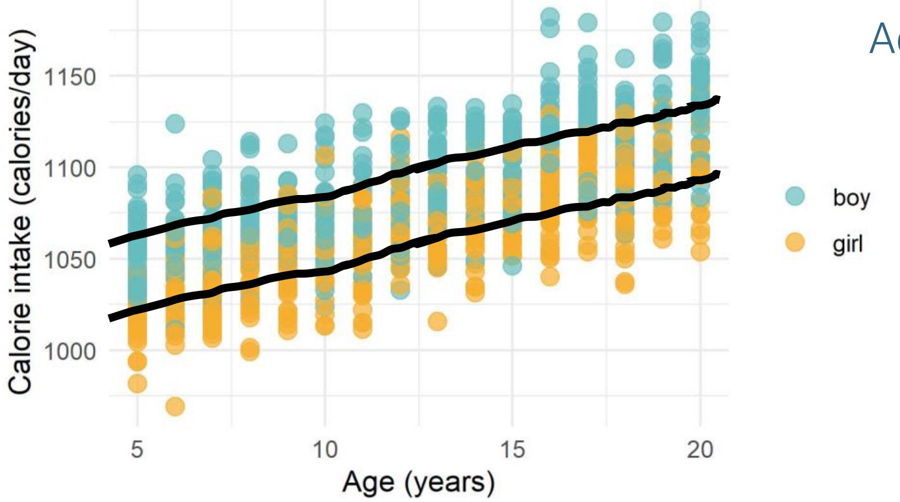
Adjusted to the number of covariates

$$\text{Cal} = \beta_0 + \beta_1\text{Gender} + \beta_2\text{Age} + \beta_3\text{Wealth} + \cdots ?$$

$$\text{Cal} = \beta_0 + \beta_1\text{Gender} + \beta_2\text{Age}$$

$$R^2{}_{\text{adj}} = 0.63$$





Gender

Age

- **Multivariate linear** model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k \text{ and}$$

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon_i$$

Multivariate linear regressions: assumptions

- **Independence**: Y values are statistically independent (via ε_i)
- **Linearity between $\boldsymbol{X}$s and $\boldsymbol{Y}$**: the means of the subpopulations of Y all lie on the same straight line defined by the regression model
- **Normality**: for each combinations of Xs there is a subpopulation of Y values; these subpopulations must be normally distributed
- **Homoscedasticity**: the variances of the subpopulations of Y are all equal and denoted by σ^2
- **Absence of co-linearity**: between covariates

To avoid redundancy across covariates

Assumptions are on the subpopulations of $\boldsymbol{Y}$ for each combinations of $\boldsymbol{X}$s

Can only be assessed via the residuals

Also need to test for linear correlations between covariates

Case study

- Blood pressure ~ individual characteristics
  - Gender
  - Age
  - Body mass index (BMI)
  - Cholesterol level
  - Smoking habits
  - Physical activity
  - Diet quality
  - Family history
  - Stress level
  - …

Biological assumptions?

You

Give me an example of a dataset I can simulate with R and use for a lecture on multivariate model building and selection that (1) relates to public health, (2) include corelated covariates (collinearity) in the model selection process, and (3) include some non-explicative variables to be dropped in the model selection process

ChatGPT

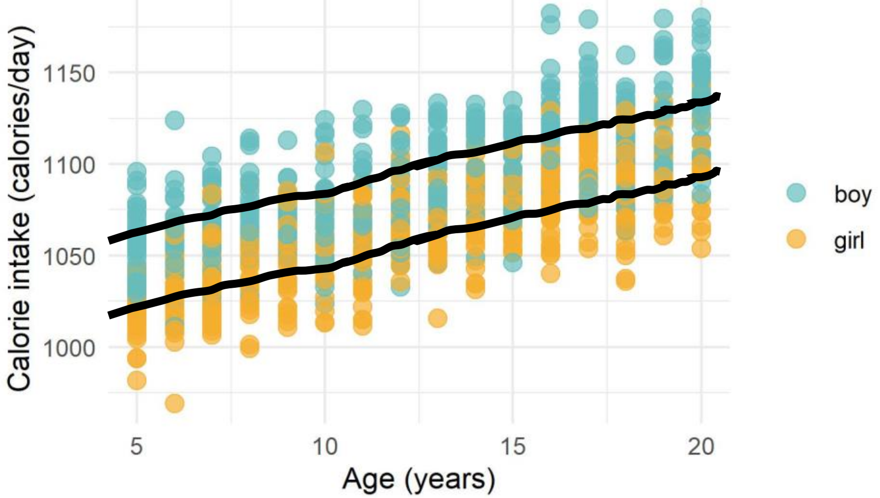
Sure! Here's an example dataset that you could simulate in R for your lecture:

Suppose we're studying factors affecting cardiovascular disease risk in a population. We collect data on various demographic, lifestyle, and health-related factors for a sample of individuals.

1. **Age**: Age of the individual.
2. **Gender**: Gender of the individual (male/female).
3. **Body Mass Index (BMI)**: Body mass index, a measure of body fat based on height and weight.
4. **Blood Pressure**: Systolic blood pressure measurement.
5. **Cholesterol Levels**: Total cholesterol levels in the blood.
6. **Smoking Status**: Smoking status of the individual (smoker/non-smoker).
7. **Physical Activity**: Number of hours of physical activity per week.
8. **Diet Quality**: A score representing the overall quality of the individual's diet.
9. **Family History**: Binary variable indicating whether the individual has a family history of cardiovascular disease.
10. **Stress Level**: Subjective measure of stress experienced by the individual.

Today's themes

- Multivariate linear models
  - 0^{th} step – Remember the biological assumptions
  - 1^{st} step – Remember of linear model assumptions
  - 2^{nd} step – Identify the most parsimonious model
  - 3^{rd} step – Make inference



Starting point: the most biologically relevant complete model

- Blood pressure ~ individual characteristics
  - Gender
  - Age
  - Body mass index (BMI)
  - Cholesterol level
  - Smoking habits
  - Physical activity
  - Diet quality
  - Family history
  - Stress level

$$Blood\ pressure = \beta_0 + \beta_1 Gender + \beta_2 Age + \cdots$$

$$+ \beta_3 Smoking \times Stress + \cdots$$

With potential interaction (effect on the slope)?

```
blood_pressure ~ age + gender + bmi + cholesterol + smoking +
physical_activity + diet_quality + family_history + stress_level
```

Multivariate linear regressions: assumptions

- **Independence**: Y values are statistically independent (via ε_i)
- **Linearity between Xs and Y**: the means of the subpopulations of Y all lie on the same straight line defined by the regression model
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On the residuals

- **Homoscedasticity**: the variances of the subpopulations of Y are all equal and denoted by σ^2
- **Absence of co-linearity**: between covariates

To avoid redundancy across covariates

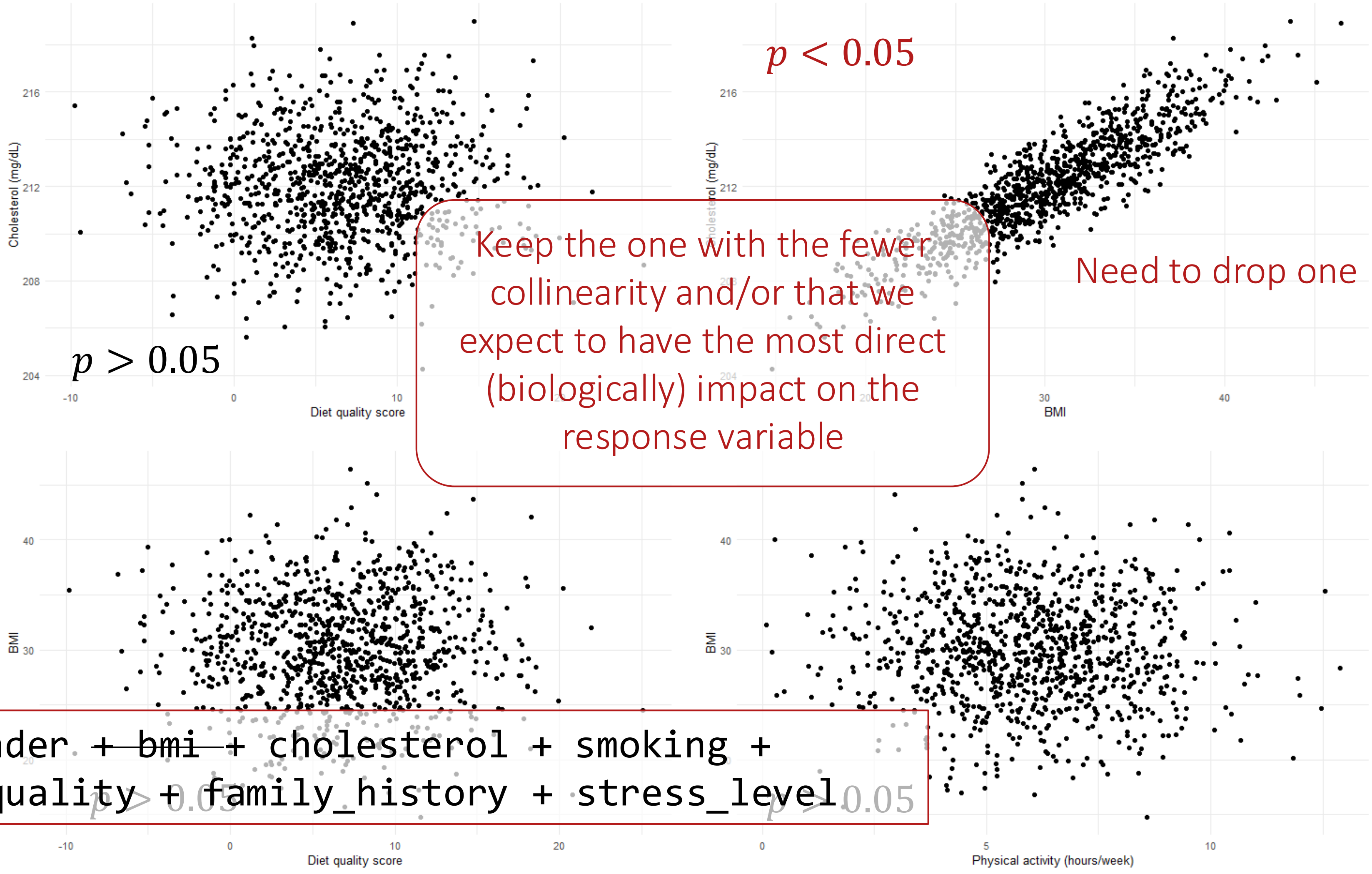
Assumptions are on the subpopulations of Y for each combinations of Xs

Can only be assessed via the residuals

Also need to test for linear correlations between covariates

Assessing collinearity between covariates (Xs)

- Pair-wise explorations of potential correlations between covariates
  - Biology-informed
    - Gender
    - Age
    - ~~Body mass index (BMI)~~
    - Cholesterol level
    - Smoking habits
    - Physical activity
    - Diet quality
    - Family history
    - Stress level
  - Visual
  - Statistical



Keep the one with the fewer collinearity and/or that we expect to have the most direct (biologically) impact on the response variable

Need to drop one

blood_pressure ~ age + gender ~~+ bmi~~ + cholesterol + smoking +
physical_activity + diet_quality + family_history + stress_level

Multivariate linear regressions: assumptions

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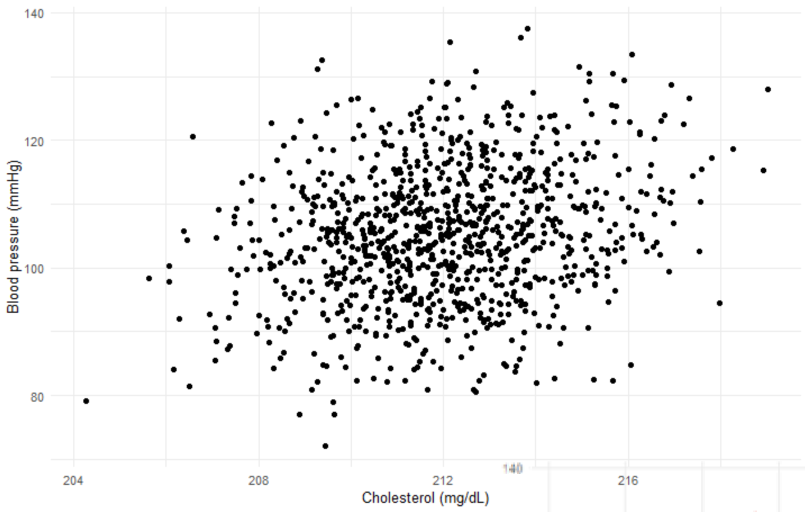
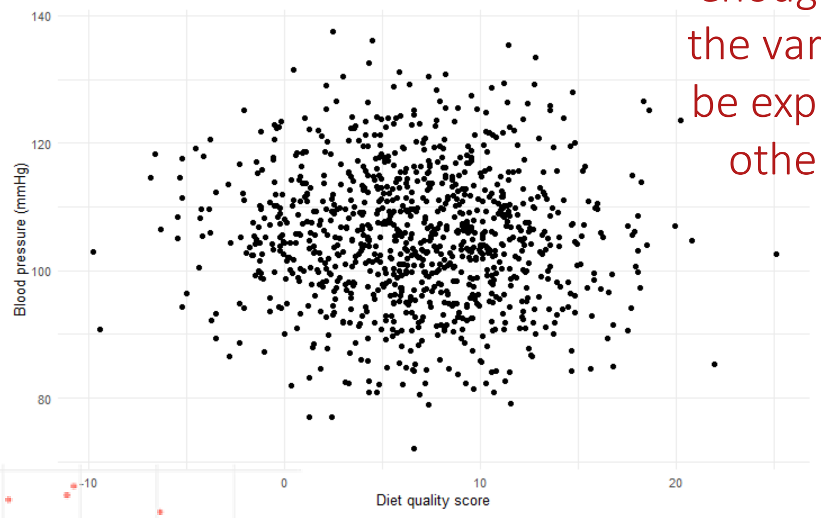
Can only be assessed via the residuals

Also need to test for linear correlations between covariates

Investigating linearity with the individual covariates

- Pair-wise explorations of the relationship between the response variable and the individual explanatory covariates
  - Visual





This is “linear enough” (as lots of the variability might be explained by the other variables)

Interactions?

Investigating linearity with the individual covariates

- Pair-wise explorations of the relationship between the response variable and the individual explanatory covariates
  - Visual
  - Variance inflation factor

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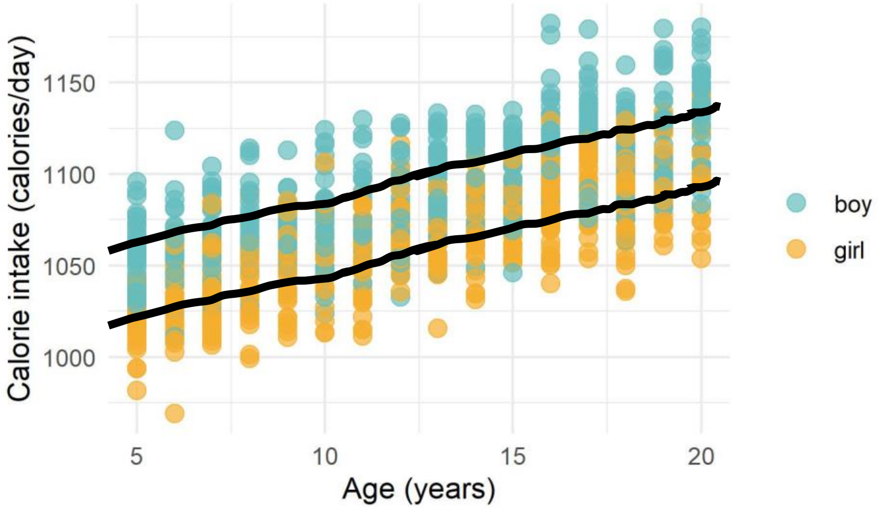

On the (un)universality of the model selection problem

- I use the **Akaike information criterion**, ~ # covariates and likelihood (of observing the data with the model parameter)
- Daniel & Cross, 2019 advise **stepwise selection**

And it goes on, and on, and on…

You
Can you list all the different methods people use for model selection?

ChatGPT
Certainly! Model selection is a crucial step in statistical modeling and regression analysis. Various methods are employed to choose the best model among a set of candidate models. Here are some common methods:

1. **Stepwise Selection:**
   - **Forward Selection:** Start with an empty model and add predictors one by one based on their significance until no additional variables improve the model significantly.
   - **Backward Elimination:** Start with a model containing all predictors and remove them one by one based on their significance until no further removal improves the model significantly.
   - **Stepwise Selection:** Combination of forward and backward selection where variables are added or removed at each step based on significance.
2. **Information Criteria:**
   - **Akaike Information Criterion (AIC):** Penalizes models based on the number of parameters and goodness of fit. Lower AIC indicates a better model.
   - **Bayesian Information Criterion (BIC):** Similar to AIC but with a stronger penalty for model complexity. It tends to select simpler models than AIC.
3. **Cross-Validation:**
   - **k-Fold Cross-Validation:** Data is divided into k subsets, and the model is trained on k-1 subsets and tested on the remaining subset. This process is repeated k times, and the average performance is used to select the best model.
   - **Leave-One-Out Cross-Validation (LO[illegible]V):** Special case of k-fold cross-validation where k equals the number of observations. Each observation serves as the validation set once.

On the (un)universality of the model selection problem

- Many approaches to model selection
- The most commonly applied varies across fields…
- But they all target the same goal:

$$R^2 = \frac{SSY - SSE}{SSY}$$

$$R^2_{\text{adj}} = 1 - \frac{SSE/_{n-\boldsymbol{k}-1}}{SSY/_{n-1}}$$

The most parsimonious model

Trade-off between explained variance (the more covariates, the better) and biological relevance and meaningfulness (the least covariates the better)

Statistical conclusions can only be made after model selection as model structure impacts effect size estimates (and p-values)

On the (un)universality of the model selection problem

- Stepwise Selection:
  - Forward Selection: Start with an empty model and add predictors one by one based on their significance until no additional variables improve the model significantly.
  - Backward Elimination: Start with a model containing all predictors and remove them one by one based on their significance until no further removal improves the model significantly.
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- Information Criteria:
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- Cross-Validation:
  - k-Fold Cross-Validation: Data is divided into k subsets, and the model is trained on k-1 subsets and tested on the remaining subset. This process is repeated k times, and the average performance is used to select the best model.
  - Leave-One-Out Cross-Validation (LOOCV): Special case of k-fold cross-validation where k equals the number of observations. Each observation serves as the validation set once.
- Regularization Methods:
  - Lasso Regression (L1 Regularization): Penalizes the absolute size of regression coefficients, forcing some coefficients to be exactly zero, effectively performing variable selection.
  - Ridge Regression (L2 Regularization): Penalizes the square of regression coefficients, shrinking coefficients towards zero but not necessarily to zero.
- Subset Selection:
  - Best Subset Selection: Fits all possible combinations of predictors and selects the best subset based on a criterion such as AIC or BIC.
  - Forward Stagewise Selection: Similar to forward selection but updates the coefficients in small steps to avoid overfitting.
- Bootstrap Resampling:
  - Bootstrap Aggregating (Bagging): Generates multiple bootstrap samples from the data and fits models to each sample. The final model is an aggregation of the individual models.
  - Random Forests: Extension of bagging where decision trees are used as base models. Random forests reduce overfitting and provide variable importance measures.
- Information-Theoretic Approaches:
  - Model Averaging: Instead of selecting a single best model, averages predictions from multiple models weighted by their likelihood or information criterion scores.
- Expert Judgment:
  - Subjective Selection: Involves the expert judgment of researchers or domain experts to select the most appropriate model based on theoretical considerations or prior knowledge.

Forward model selection

- Start with a univariate model (with the likely most important covariate) and add covariates one by one, dropping them if they do not turn out significant (effect size different from 0)

Potential full model:

```
blood_pressure ~ age + gender ~~+ bmi~~ + cholesterol + smoking +
physical_activity + diet_quality + family_history + stress_level
```

Forward selection:

blood_pressure ~ cholesterol $\beta_1 = 1.1 \pm 0.3$

blood_pressure ~ cholesterol ~~+ stress_level~~ $\beta_1 = 1.1 \pm 0.3$ $\beta_2 = 0.2 \pm 0.3$

blood_pressure ~ cholesterol ~~+ physical_activity~~ $\beta_1 = 1.1 \pm 0.3$ $\beta_2 = -0.3 \pm 0.3$

blood_pressure ~ cholesterol + age $\beta_1 = 1.2 \pm 0.3$ $\beta_2 = 0.1 \pm 0.0$

blood_pressure ~ cholesterol + age + gender $\beta_1 = 1.2 \pm 0.3$ $\beta_2 = 0.1 \pm 0.0$ $\beta_3 = 0.2 \pm 1.2$

Today's themes

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  - 0th step – Remember the biological assumptions
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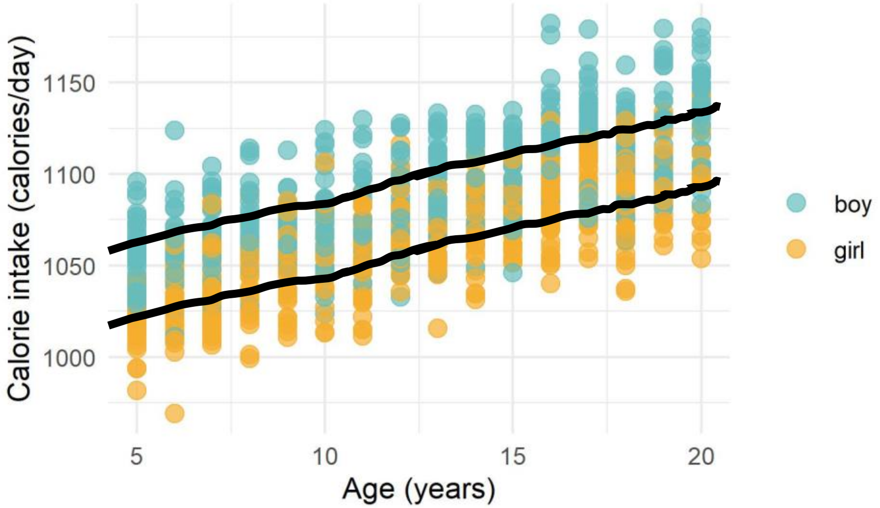

Inference from multivariate models

- Check the assumptions on the residuals (see previous lecture)

- Remember the hypothesizes
  - H_0: all regression coefficients are equal to zero or R-squared is equal to zero
  - H_1: at least one regression coefficients is not equal to zero or R-squared is not equal to zero

- Report the significantly correlated covariates, their effect size and the corresponding p-value (as for a univariate model)

- Non-significantly correlated variables can be kept to “adjust” the model (e.g., gender-adjusted model)

Linear models entirely rely on slope estimates

- Remember the hypothesizes
  - H_0: all regression coefficients are equal to zero or R-squared is equal to zero
  - H_1: at least one regression coefficients is not equal to zero or R-squared is not equal to zero
- Conclusions are based on the slope estimate…
- A slope will always be estimated (i.e., you can always find the "best" line), but it's our job to assess if that estimate is legit (i.e., sometime the "best" line is still a bad line)

Back to the Assumptions

Multivariate linear regressions: <u>assumptions</u>, for the 50th time

- **Independence**: Y values are statistically independent (via ε_i)
- **Linearity between $\boldsymbol{X}$s and $\boldsymbol{Y}$**: the means of the subpopulations of Y all lie on the same straight line defined by
- **Normality**: <u>for each combinations of Xs</u> there is a subpopulation of Y values; these subpopulations must be normally distributed
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Study design

Step 0 – **Biological assumption**

Step 1 – Visual exploration of the **raw data**

Step 2 – Visual exploration of the model residuals

Checking the residuals…

- 1st step is to get the residuals (fitting the model)
- Blood pressure ~ individual characteristics
    - Gender
    - Age
    - Body mass index (BMI)
    - Cholesterol level
    - Smoking habits
    - Physical activity
    - Diet quality
    - Family history
    - Stress level
- Forward model selection: `blood_pressure ~ age + cholesterol`

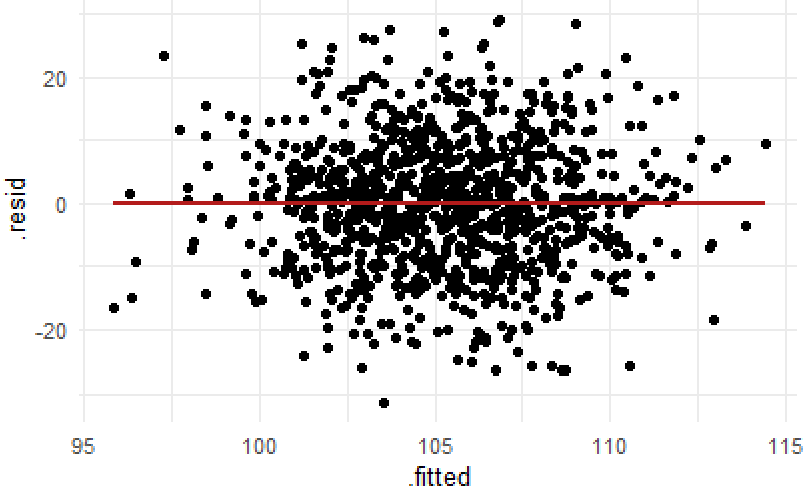
$$Blood\ pressure_i = \beta_0 + \beta_1 Age_i + \beta_2 Cholesterol_i + \varepsilon_i$$

$$\beta_0 = -149 \pm 60$$
$$\beta_1 = 0.15 \pm 0.06$$
$$\beta_2 = 1.16 \pm 0.27$$

Assessing linearity from the residuals

- Visual

`blood_pressure ~ age + cholesterol`



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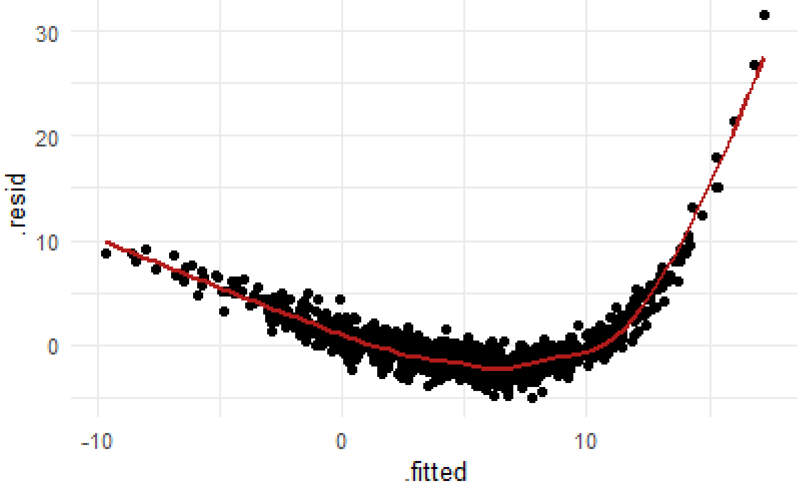
`y ~ x` with $y_i = 3e^x + \varepsilon_i$

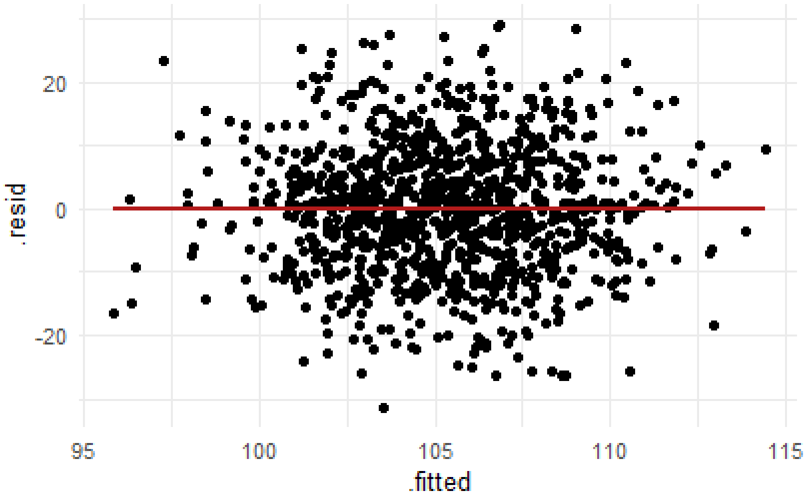


=(

Assessing linearity from the residuals

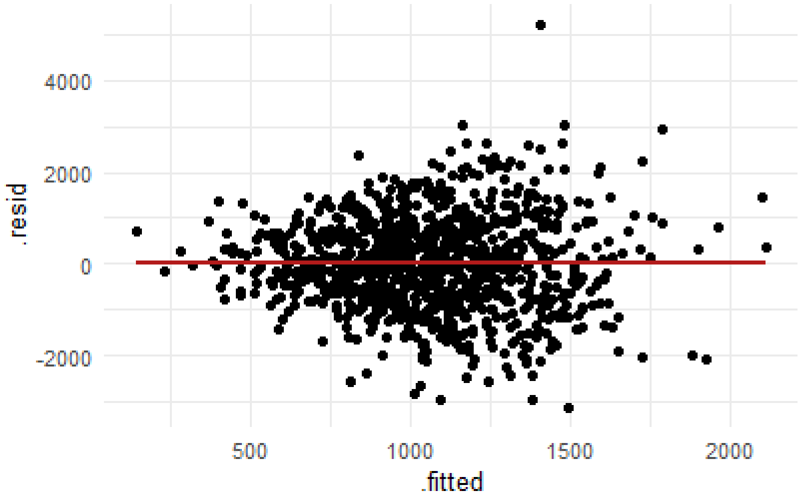
- Visual

`blood_pressure ~ age + cholesterol`



=)

`y ~ x` with $y_i = 2x + \varepsilon_i$ and $\varepsilon \sim N(0, 2x)$



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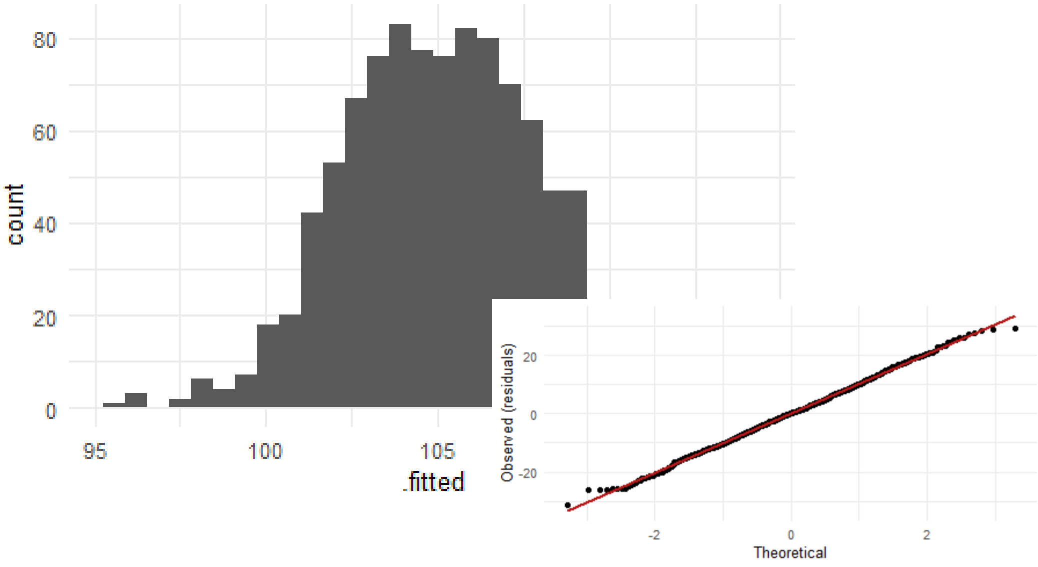
Assessing normality of the residuals

- Visual
- Statistical? The Shapiro-Wilk test

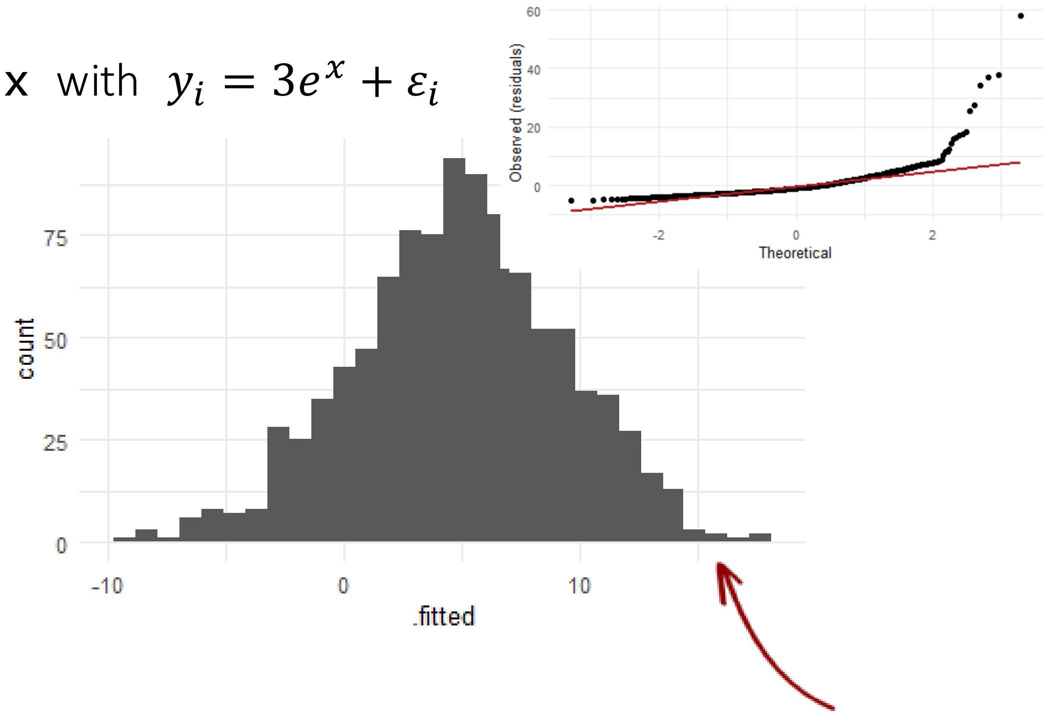
`blood_pressure ~ age + cholesterol`



=)

$p = 0.6$ =)

`y ~ x` with $y_i = 3e^{x} + \varepsilon_i$



=) ?

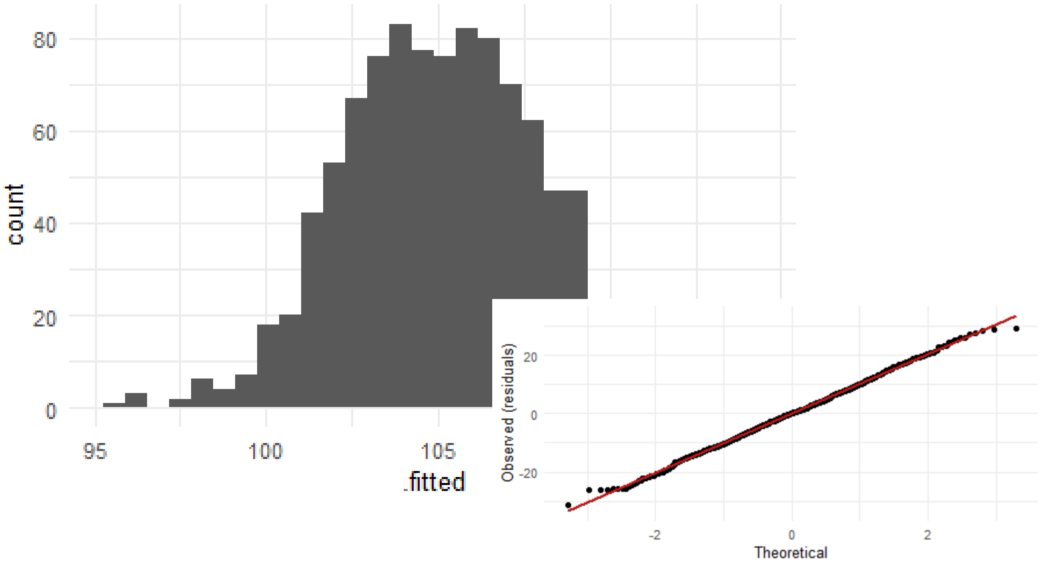
Extreme values

$p < 0.05$ =(

Assessing normality of the residuals

- Visual
- Statistical? The Shapiro-Wilk test

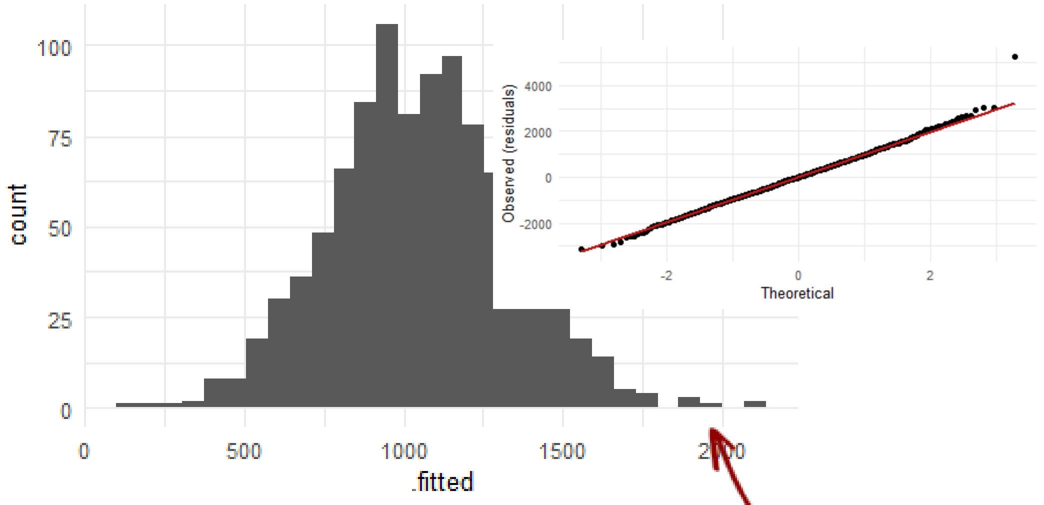
`blood_pressure ~ age + cholesterol`



=)

$p = 0.6$ =)

`y ~ x with` $y_i = 2x + \varepsilon_i$ `and` $\varepsilon \sim N(0, 2x)$



Extreme values

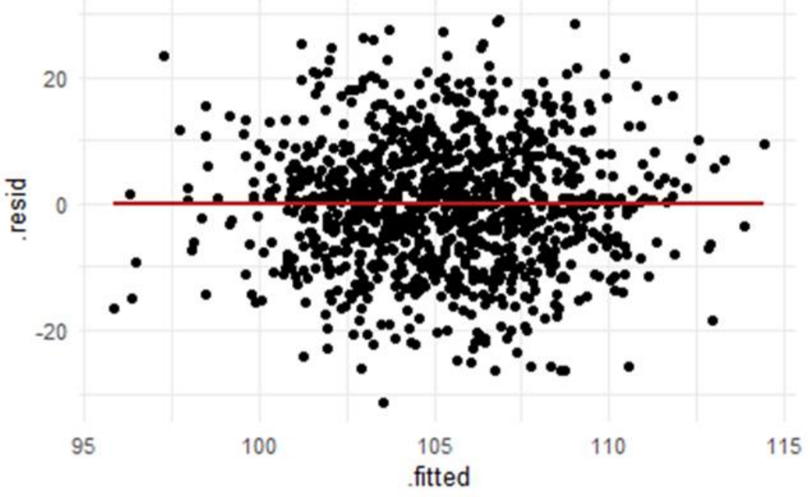
=) ?

$p < 0.05$ =(

Assessing homoscedasticity of the residuals

- Visual
- Statistical? The Breusch-Pagan test

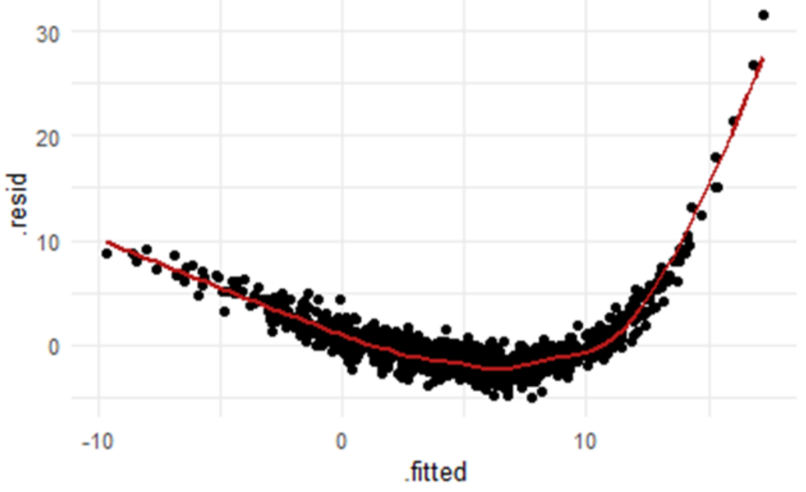
`blood_pressure ~ age + cholesterol`



=)

$$p = 0.001$$ =/

`y ~ x` with $y_i = 3e^x + \varepsilon_i$



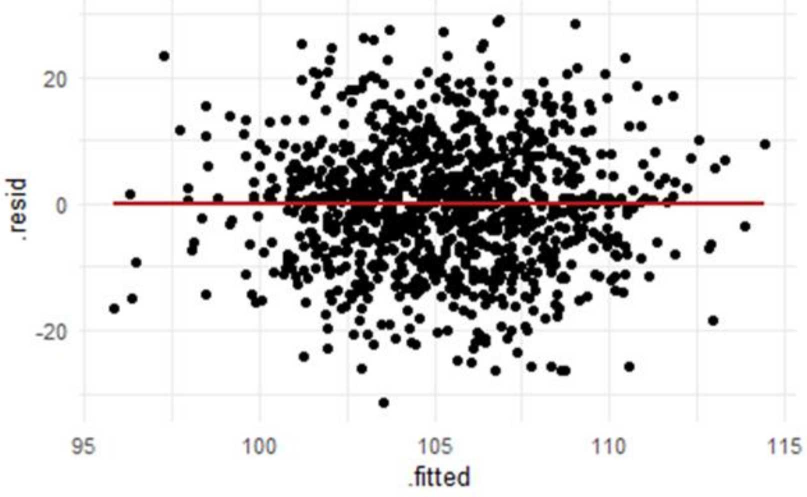
=(

$$p < 0.001$$ =(

Assessing homoscedasticity of the residuals

- Visual
- Statistical? The Breusch-Pagan test

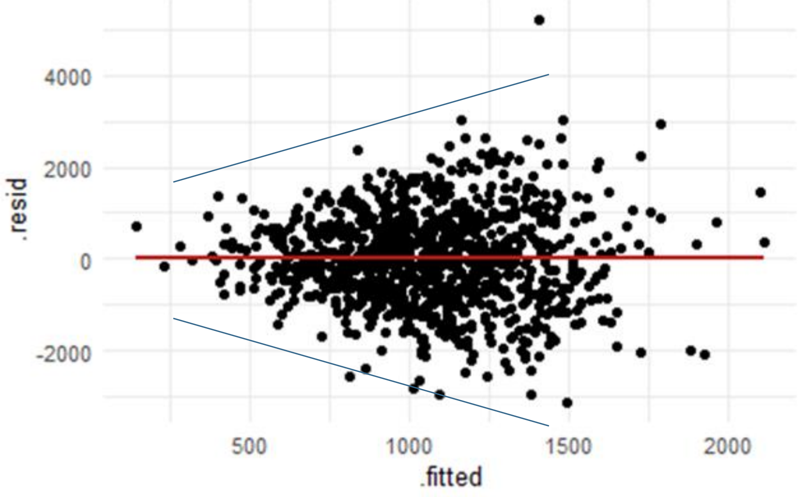
`blood_pressure ~ age + cholesterol`



=)

$p = 0.001$ =/

`y ~ x` with $y_i = 2x + \varepsilon_i$ and $\varepsilon \sim N(0, 2x)$



=(

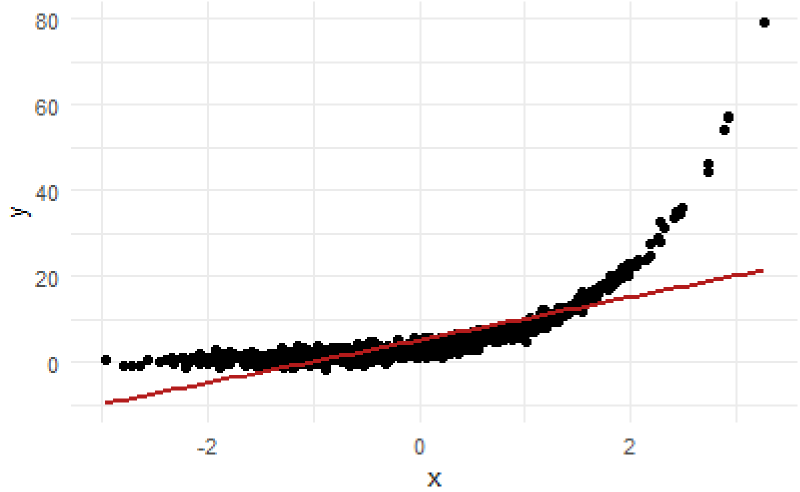
$p < 0.001$ =(

On the importance of checking assumptions before conclusion

- Datasets violating the assumptions for linear models can lead to outputs suggesting significant associations between co-variates
- Plotting the raw data is always the first step, plotting the residuals is also mandatory

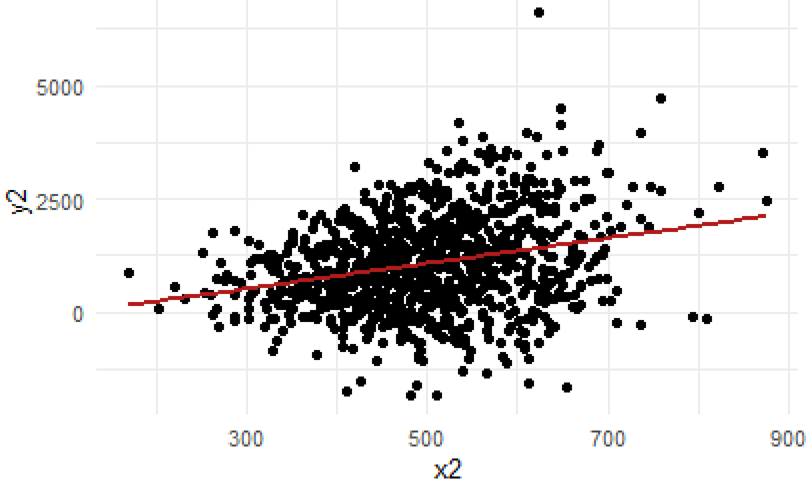
y ~ x with $y_i = 3e^x + \varepsilon_i$

$\beta_1 = 5.01 \pm 0.26, p < 0.05$
$R^2 = 0.5$



y ~ x with $y_i = 2x + \varepsilon_i$ and $\varepsilon \sim N(0, 2x)$

$\beta_1 = 2.79 \pm 0.61, p < 0.05$
$R^2 = 0.07$



Assessing collinearity

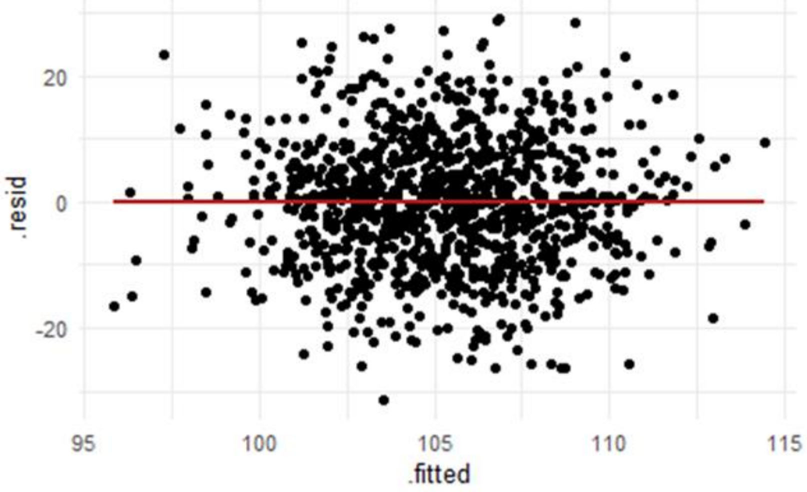
- Before model fitting: pair-wise correlation between covariates
  - Visual
  - Statistical
- After model fitting: the variance inflation factor

Measure of collinearity

$$VIF_j = \frac{1}{1-R_j^2}$$

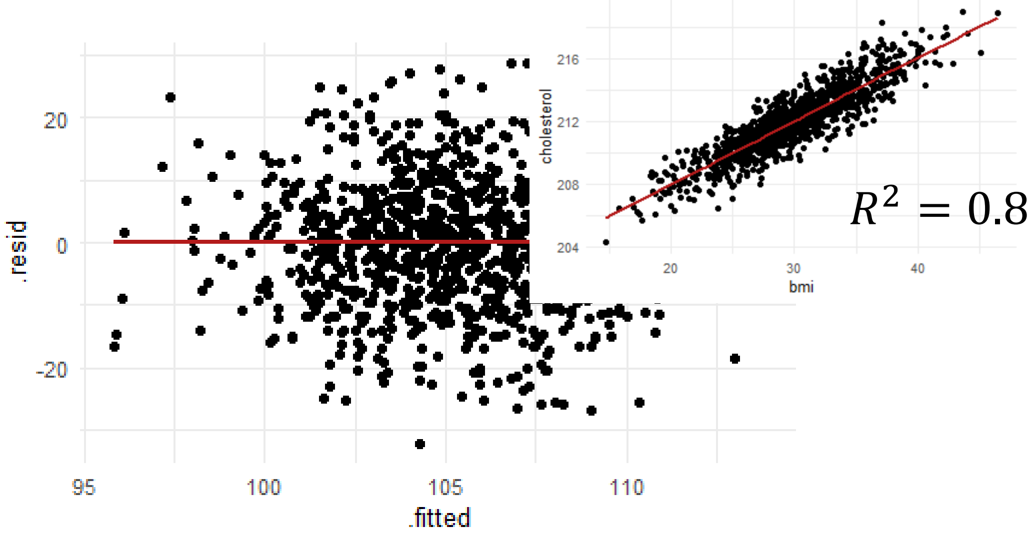
with R_j the unadj. determination coeff. of the regression of variable j over the other variables

`blood_pressure ~ age + cholesterol`



$VIF_{Cholesterol} = 1.004$
$VIF_{Age} = 1.004$

`blood_pressure ~ age + bmi + cholesterol`



$\boldsymbol{VIF_{Cholesterol} = 5.001}$
$\boldsymbol{VIF_{BMI} = 4.998}$
$VIF_{Age} = 1.004$

Assessing collinearity

- Before model fitting: pair-wise correlation between covariates
  - Visual
  - Statistical
- After model fitting: the variance inflation factor

Measure of collinearity

$$VIF_j = \frac{1}{1-R_j^2}$$

with R_j the unadj. determination coeff. of the regression of variable j over the other variables

`blood_pressure ~ age + cholesterol`

$$\beta_0 = -149 \pm 60$$
$$\beta_1 = 0.15 \pm 0.06$$
$$\beta_2 = 1.16 \pm 0.27$$

`blood_pressure ~ age + cholesterol + bmi`

$$\beta_0 = -149 \pm 60$$
$$\beta_1 = 0.15 \pm 0.06$$
$$\boldsymbol{\beta_2 = 0.76 \pm 0.63}$$
$$\boldsymbol{\beta_3 = 0.20 \pm 0.28}$$

A Note on Interactions

Multivariate linear regressions: formalization

- **Multivariate linear** model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$ and

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon_i$$

Those are additive infects…

- With interactions $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_{12} X_1 \times X_2 + \cdots$

The effect of X_1 changes with X_2 (and vise-versa)

What to do when assumptions are violated?

- Independence
  - Paired tests, hierarchical/mixed models, redesign the sampling protocol…
- Linearity between $\boldsymbol{X}$s and $\boldsymbol{Y}$
  - Variable transformation, non-linear models…
- Normality
  - Non-parametric tests…
- Homoscedasticity
  - Generalized models (apply a function linking the variance to the covariates, usually goes together with variable transformation)
- Absence of co-linearity
  - Remove redundant covariates

Model Checks

A parsimonious and biologically relevant model

Trade-off between explained variance (the more covariates, the better) and biological relevance and meaningfulness (the least covariates the better)

Statistical conclusions can only be made after model selection as model structure impacts effect size estimates (and p-values)

- Check the effect sizes (correlations are meaningful?)
- Check the determination coefficient (model is informative?)
- A few more things…

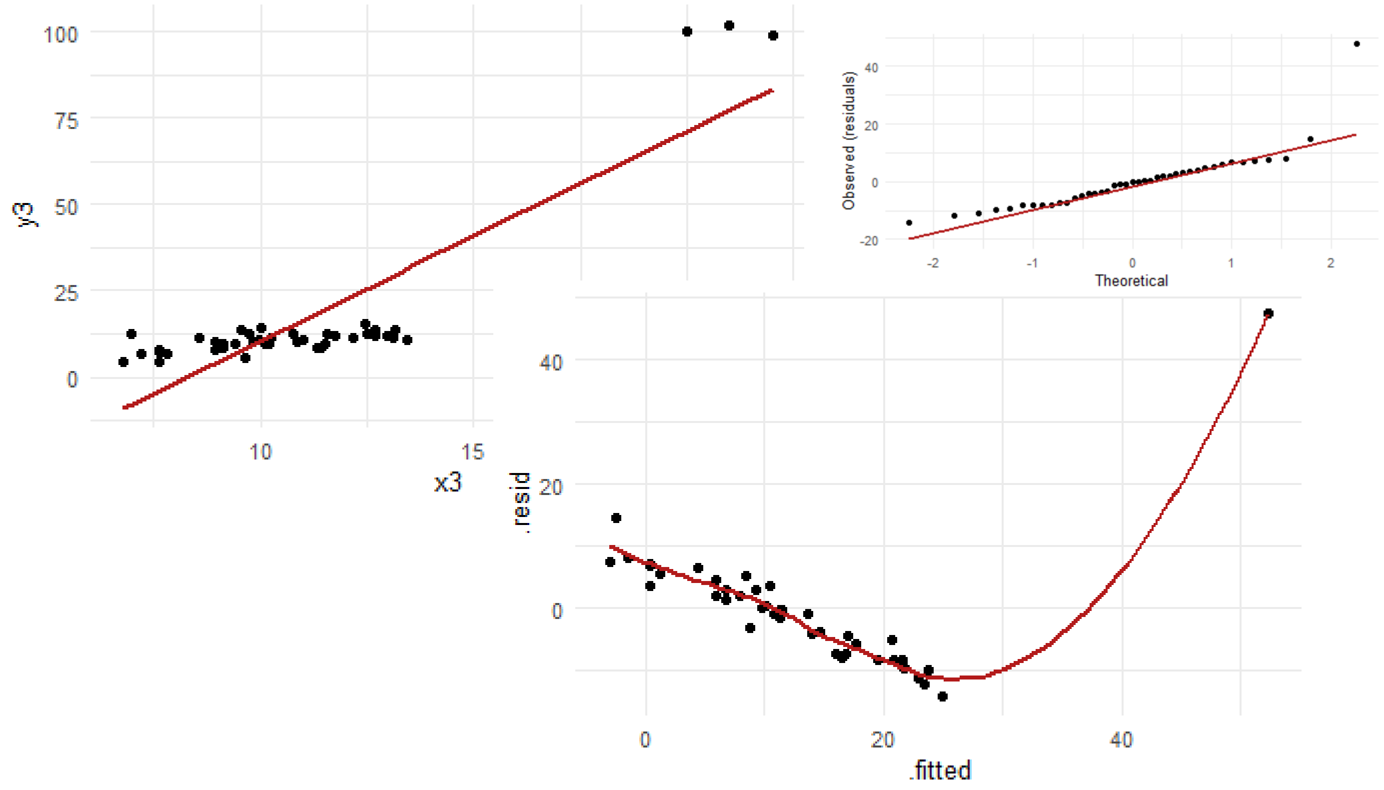
$$D_i = \frac{\sum_{j=1}^{n}(\hat{y}_j - \hat{y}_{j(i)})}{k\ MSE}$$

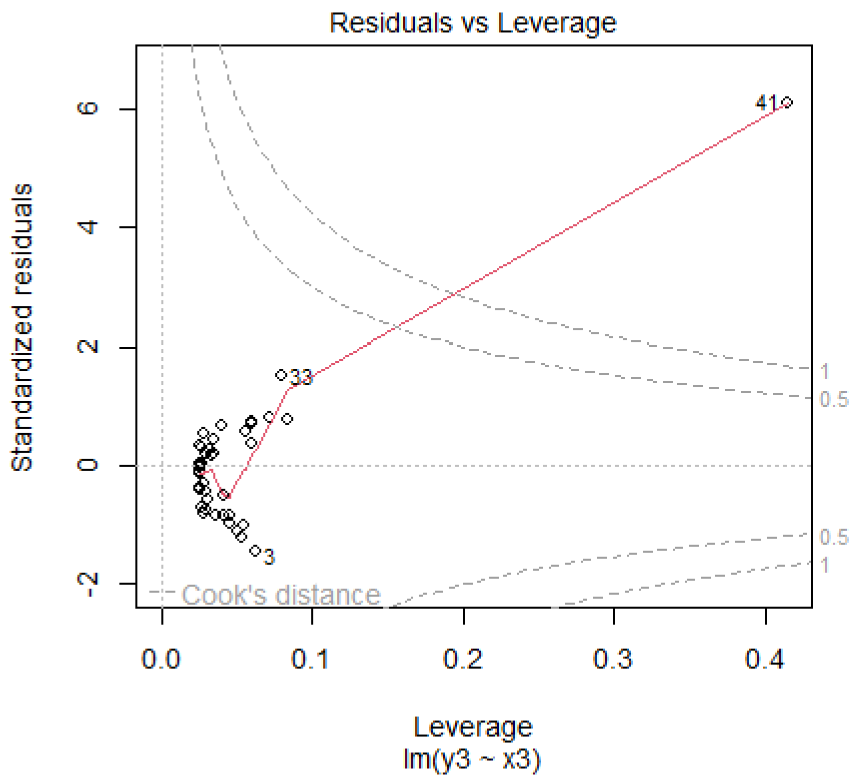
with MSE the mean square error

Not driven by a few data points

- Leverage and outliers
  - Linear regressions use means and means are sensitive to outliers and extreme values
- Cook's distance

Measure of the changes in model outputs
when removing individual data points





A parsimonious and biologically relevant model

Trade-off between explained variance (the more covariates, the better) and biological relevance and meaningfulness (the least covariates the better)

Statistical conclusions can only be made after model selection as model structure impacts effect size estimates (and p-values)

- Check the effect sizes (correlations are meaningful?)
- Check the determination coefficient (model is informative?)
- Have a look at individual points and how they could impact the conclusions

Before Next Time (Friday March 27, in LH4)

- Practice: **Homework #09** due by **Thursday 5 PM**
- Use the Guiding Quiz #09 if needed
- Check you are ready for next class: **Knowledge Check #09** due by **Friday 11 AM**

Need Help?

- Come to VTPEH 6108: Public Health Data Lab
- Come to office hours
  - **Tuesdays, 09:00-10:00 AM**
  - **Wednesdays, 05:00-06:00 PM**
  - If you cannot make it at these times, please contact the course assistant of your choice by email
  - in S1-122, unless indicated otherwise on Canvas